

# Operational Amplifier Data Collection

Table 1: Slew Rate Calculated and Measured Data

|                           | <i>LM741</i> | <i>TL071</i> | <i>MCP6002</i> |
|---------------------------|--------------|--------------|----------------|
| <i>Data Specification</i> |              |              |                |
| <i>Measured</i>           |              |              |                |

Table 2: Differential Amplifier Comparison: Calculated and Measured Data

|                            |                 |                   |                 |                   |             |
|----------------------------|-----------------|-------------------|-----------------|-------------------|-------------|
| <i>Transistor Diff-Amp</i> | $\Delta V_{CE}$ | $\Delta V_{CCCB}$ | $\Delta V_{CM}$ | $\Delta V_{Diff}$ | $CMRR_{dB}$ |
| <i>Measured</i>            |                 |                   |                 |                   |             |
| <i>TL071 Op Amp</i>        | $\Delta V-$     | $\Delta V+$       | $\Delta V_{CM}$ | $\Delta V_{Diff}$ | $CMRR_{dB}$ |
| <i>Calculated</i>          |                 |                   |                 |                   |             |
| <i>Measured</i>            |                 |                   |                 |                   |             |

Table 3: Single-Stage Op-Amp Circuit Data

|                     |            |            |            |              |            |
|---------------------|------------|------------|------------|--------------|------------|
| <i>Single-Stage</i> | <i>R1</i>  | <i>R2</i>  | <i>R3</i>  | <i>RL</i>    |            |
| <i>Calculated</i>   |            |            |            |              |            |
| <i>Measured</i>     |            |            |            |              |            |
| <i>Single-Stage</i> | <i>VCC</i> | <i>VEE</i> | <i>VIN</i> | <i>VOU</i> T | $\Delta V$ |
| <i>Calculated</i>   |            |            |            |              |            |
| <i>Measured</i>     |            |            |            |              |            |

Table 4: Two-Stage Op-Amp Circuit Data

|                   |                 |                 |                 |                 |                    |
|-------------------|-----------------|-----------------|-----------------|-----------------|--------------------|
| <i>Two-Stage</i>  | <i>R1</i>       | <i>R2</i>       | <i>R3</i>       | <i>R4</i>       | <i>R5</i>          |
| <i>Calculated</i> |                 |                 |                 |                 |                    |
| <i>Measured</i>   |                 |                 |                 |                 |                    |
| <i>Two-Stage</i>  | <i>R6</i>       | <i>RL</i>       | <i>VCC</i>      | <i>VEE</i>      | <i>VIN</i>         |
| <i>Calculated</i> |                 |                 |                 |                 |                    |
| <i>Measured</i>   |                 |                 |                 |                 |                    |
| <i>Two-Stage</i>  | <i>VOU</i> T_U1 | <i>VOU</i> T_U2 | $\Delta V_{U1}$ | $\Delta V_{U2}$ | $\Delta V_{Total}$ |
| <i>Calculated</i> |                 |                 |                 |                 |                    |
| <i>Measured</i>   |                 |                 |                 |                 |                    |

Table 5: Three-Stage Op-Amp Circuit Data

|                    |                |                |                |                   |               |
|--------------------|----------------|----------------|----------------|-------------------|---------------|
| <i>Three-Stage</i> | $R1$           | $R2$           | $R3$           | $R4$              | $R5$          |
| <i>Calculated</i>  |                |                |                |                   |               |
| <i>Measured</i>    |                |                |                |                   |               |
| <i>Three-Stage</i> | $R6$           | $R7$           | $R8$           | $R9$              | $RL$          |
| <i>Calculated</i>  |                |                |                |                   |               |
| <i>Measured</i>    |                |                |                |                   |               |
| <i>Three-Stage</i> | $VCC$          | $VEE$          | $V_{OUT\_U1}$  | $V_{OUT\_U2}$     | $V_{OUT\_U3}$ |
| <i>Calculated</i>  |                |                |                |                   |               |
| <i>Measured</i>    |                |                |                |                   |               |
| <i>Two-Stage</i>   | $\Delta V\_U1$ | $\Delta V\_U2$ | $\Delta V\_U3$ | $\Delta V\_Total$ |               |
| <i>Calculated</i>  |                |                |                |                   |               |
| <i>Measured</i>    |                |                |                |                   |               |

Table 6: Op-Amp multi-stage Frequency vs. Gain Data

|                                     | <i>Single – Stage</i> | <i>Two – Stage</i> | <i>Three – Stage</i> |
|-------------------------------------|-----------------------|--------------------|----------------------|
| $\Delta V\_Total$ <i>Calculated</i> |                       |                    |                      |
| $\Delta V\_Total$ <i>Measured</i>   |                       |                    |                      |
| $FC_{LOW}$ <i>Calculated</i>        |                       |                    |                      |
| $FC_{LOW}$ <i>Measured</i>          |                       |                    |                      |
| $FC_{HIGH}$ <i>Calculated</i>       |                       |                    |                      |
| $FC_{HIGH}$ <i>Measured</i>         |                       |                    |                      |